# ORF363/COS323 Optimization – Part 1 (Lec3.pdf)

Ioannis Akrotirianakis

# Topics & Logistics

- Basic notation & terminology
- Existence of optimal solutions
- Unconstrained Optimization
  - 2 examples

# Notation & Terminology

minimize & decision vars

Objective function

$$\min_{x \in R^n} \quad f(x)$$

Subject to

Set of constraints or Feasible region

$$\text{s.t.} \qquad x \in \Omega$$

# Notation & Terminology

minimize & decision vars

$$\min_{x \in R^n} \quad f(x)$$

Objective function

Subject to

$$\text{s.t.} \qquad x \in \Omega$$

Set of constraints or Feasible region

Constraint set or Feasible region

$$\Omega = \{x \in R^n \mid \quad g_i(x) \leq 0\,, i = 1,2, \dots m \quad and \quad h_j(x) = 0, j = 1,2, \dots, r\,\}$$

where: $g_i: R^n \to R$ and $h_j: R^n \to R$, for all $i, j$

# Notation & Terminology

minimize & decision vars

$$\min_{x \in R^n} \quad f(x)$$

Objective function

Subject to

$$\text{s.t.} \qquad x \in \Omega$$

Set of constraints or Feasible region

Constraint set or Feasible region

$$\Omega = \{x \in R^n \mid \; g_i(x) \leq 0\,, i = 1,2, \dots m \quad and \quad h_j(x) = 0, j = 1,2, \dots, r\,\}$$

where: $g_i: R^n \to R$ and $h_j: R^n \to R$, for all $i, j$

$$\min_{x \in R^n} \quad f(x)$$

$$\text{s.t.} \quad \mathrm{g_i}(x) \leq 0, j = 1, \dots m$$

$$h_j(x) = 0, j = 1, \dots r$$

$$\min_{x \in R^n} \quad f(x)$$

$$\text{s.t.} \quad \mathbf{g}(x) \leq 0, \qquad (\boldsymbol{g}: R^n \to R^m)$$

$$\boldsymbol{h}(x) = 0 \qquad (\boldsymbol{h}: R^n \to R^r)$$

# Notation & Terminology

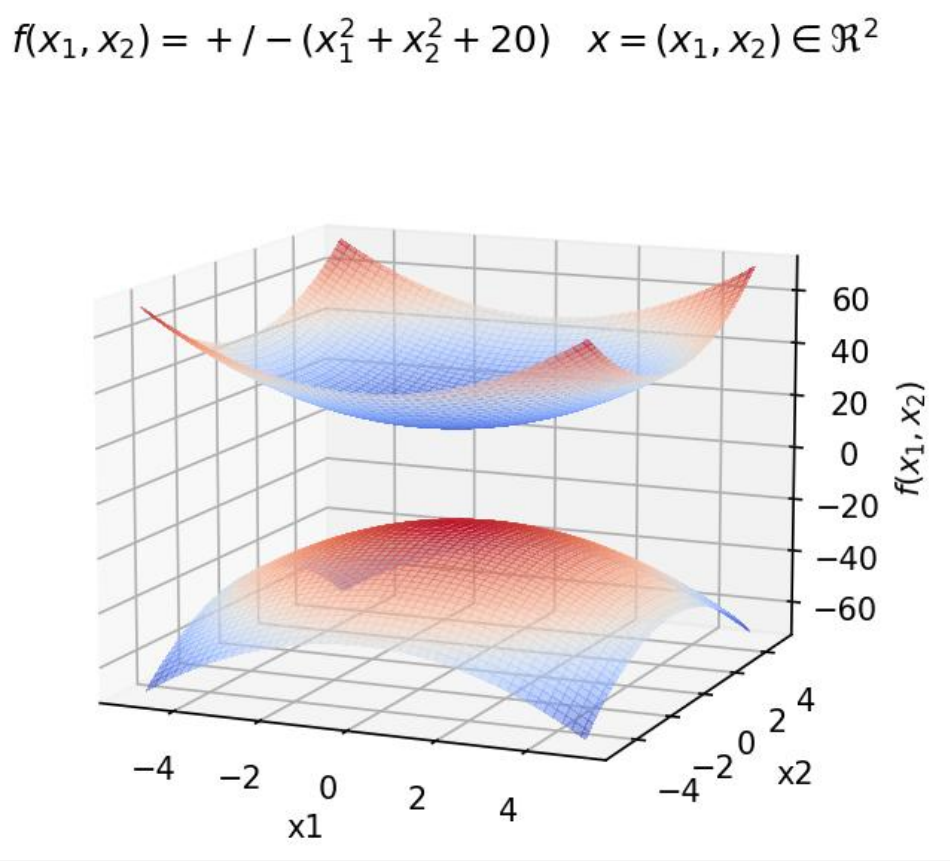
- **Feasible point**: every $x \in \Omega$ *(every decision that satisfies the constraints)*
- $x^*$ is called an **Optimal Solution** iff:
  - $\boldsymbol{x}^* \in \boldsymbol{\Omega}$ and $\boldsymbol{f}(\boldsymbol{x}^*) \leq \boldsymbol{f}(\boldsymbol{x}), \forall \boldsymbol{x} \in \boldsymbol{\Omega}$
  - $x^*$ is also called: solution or global solution or $\mathbf{arg}\,\boldsymbol{min}\ (\boldsymbol{f})$
- If $x^*$ exists, then $f(x^*)$ is called the **Optimal Value**

# Minimization -vs- Maximization

$$\min_{x \in R^n} \; f(x) \qquad \text{s.t.} \quad x \in \Omega$$

$$\leftrightarrow$$

$$-\max_{x \in R^n} \; -f(x) \qquad \text{s.t.} \quad x \in \Omega$$




**Remarks**:

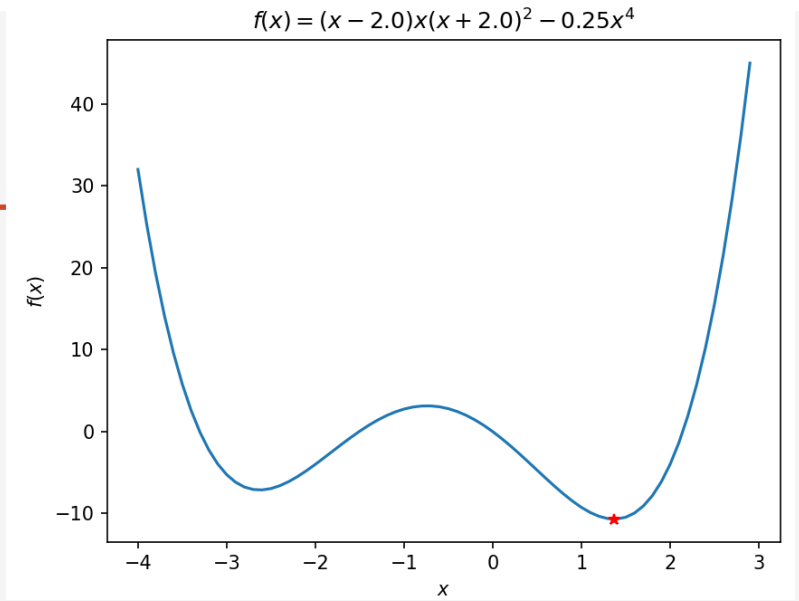
1. The optimal solution $x^*$ does not change.

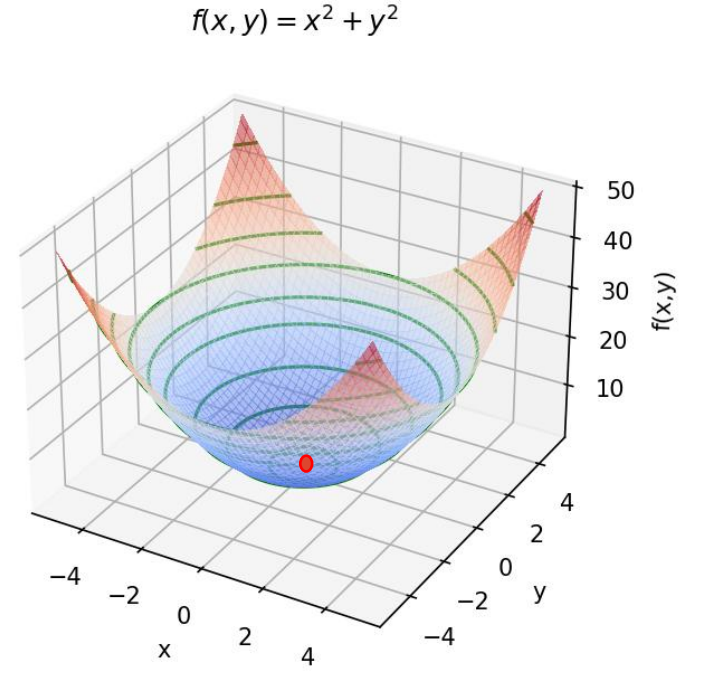
2. Only the optimal value changes.

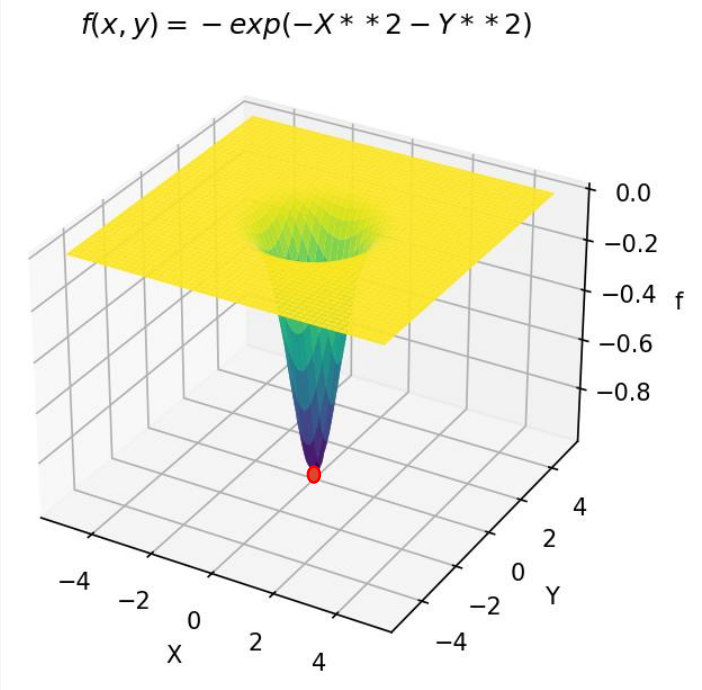
$$x^* = (0,0)^T$$
$$f(x^*) = 20$$
$$-f(x^*) = -20$$

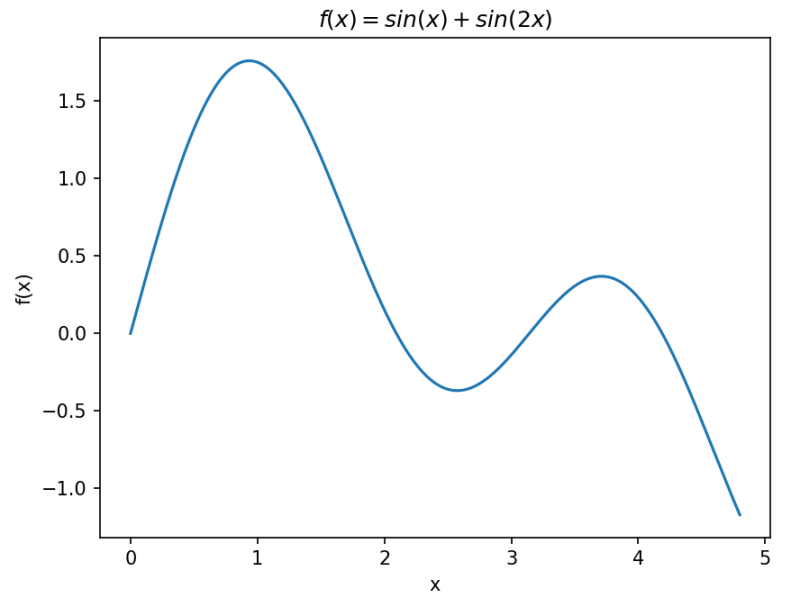
# Characteristics of optimal solutions and values

## #1. The optimal solution **exists and is unique**



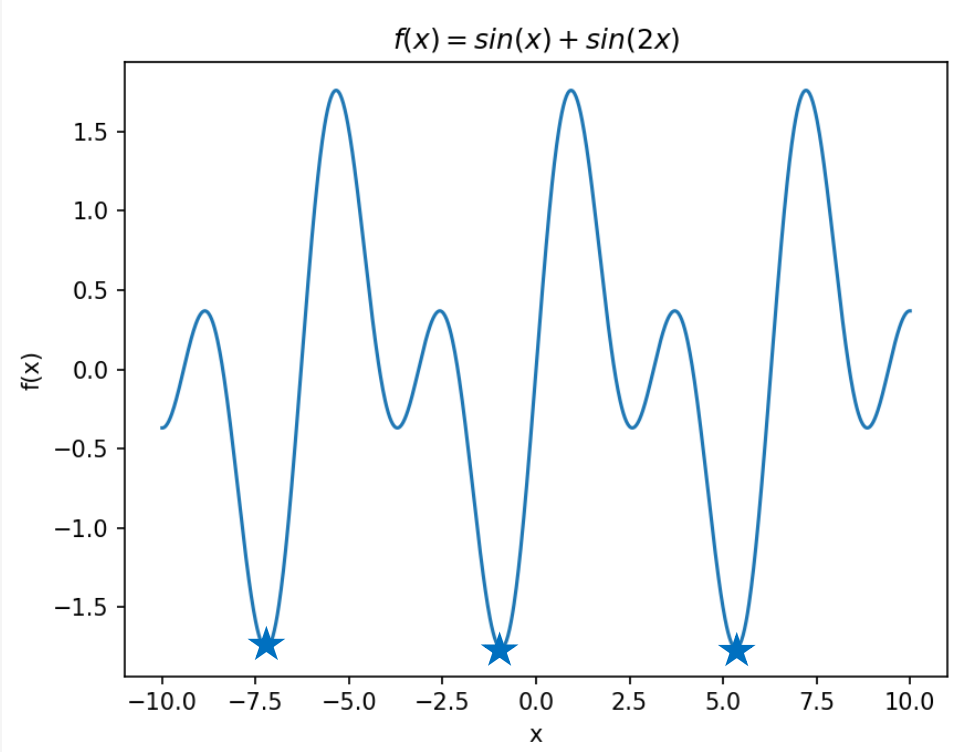



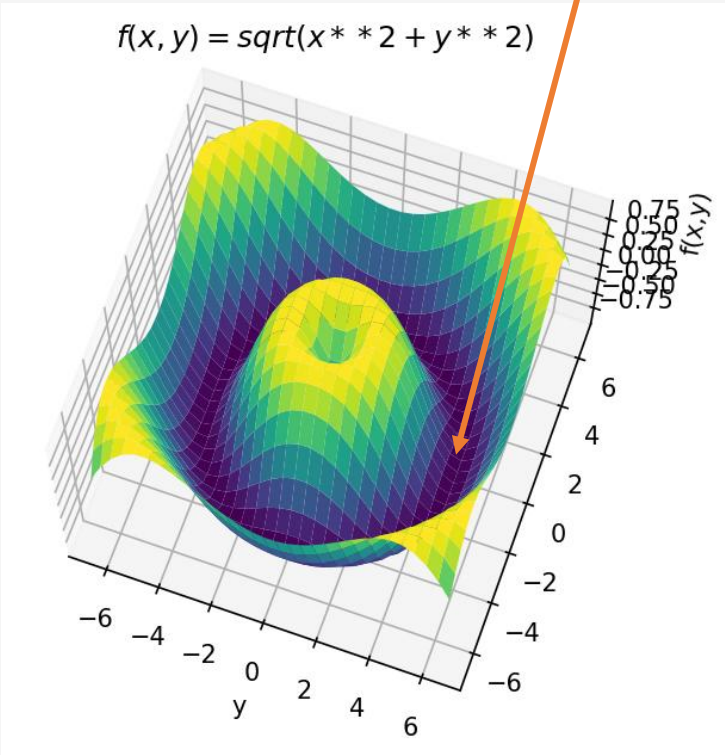

*Remember we minimize*

# Characteristics of optimal solutions and values

2. The optimal solution **exists but is not unique**

All points at the deepest blue are optimal solutions (**infinite number** of them)




**But the optimal value is always unique**

*Remember we minimize*

# Characteristics of optimal solutions and values

3. The optimal solution **may not exist**

   a. If the feasible region is empty $\Omega = \emptyset$ (**infeasible problem**)

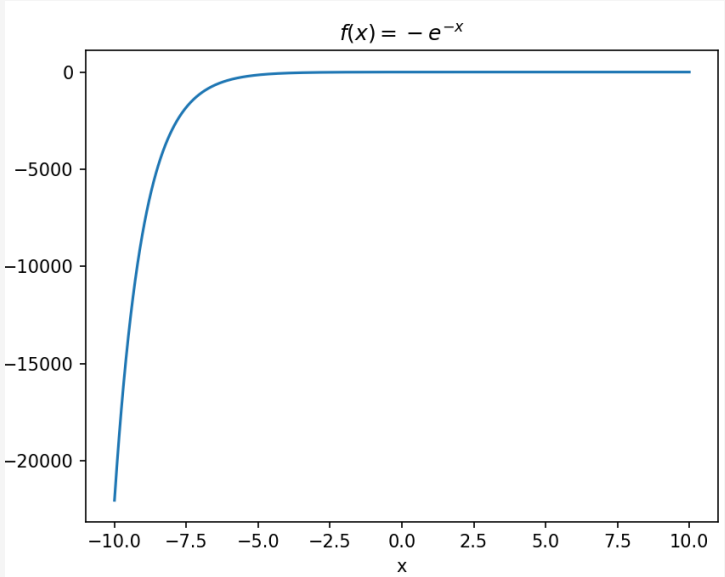
   **b.** If the function is **unbounded below**

   ~ but we say the *optimal value is* $-\infty$

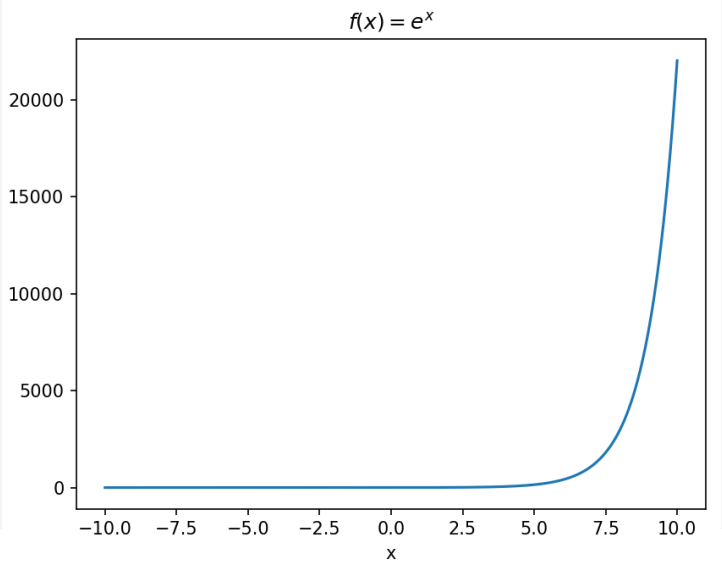
   **c.** The optimal value is **not attained**

   ~ but we say the *optimal value is the largest lower bound of* $f$ (also known as the *infimum*). For our example (c), we say the optimal value is 0 (but there is no optimal solution)

*Remember we minimize*




**b.**




**c.**

# Summary of Optimality Characterizations

- 1. May exist & be unique
- 2. May exist and & be not unique
- 3. May not exist
  - 3.1 infeasible problem
  - 3.2 Unbounded problem
  - 3.3 Is not attained

# Weierstrass theorem → guarantees existence of $x^*$

- **Theorem**: If $f$ is continuous function and $\Omega$ is compact set (i.e., closed & bounded), then **an optimal solution to** $\boxed{\begin{array}{ll} \min\limits_{x \in R^n} & f(x) \\ \text{s.t.} & x \in \Omega \end{array}}$ **always exist**

- A set S is called **closed set** iff you take any infinite sequence of points in the set and the limit of that sequence is also in the set. Example??

- A set S is called **bounded set** iff you can enclose it in a ball of finite radius. Example??

# Unconstrained Optimization

- We have $\Omega = R^n$

$$\min_{x \in R^n} \; f(x)$$

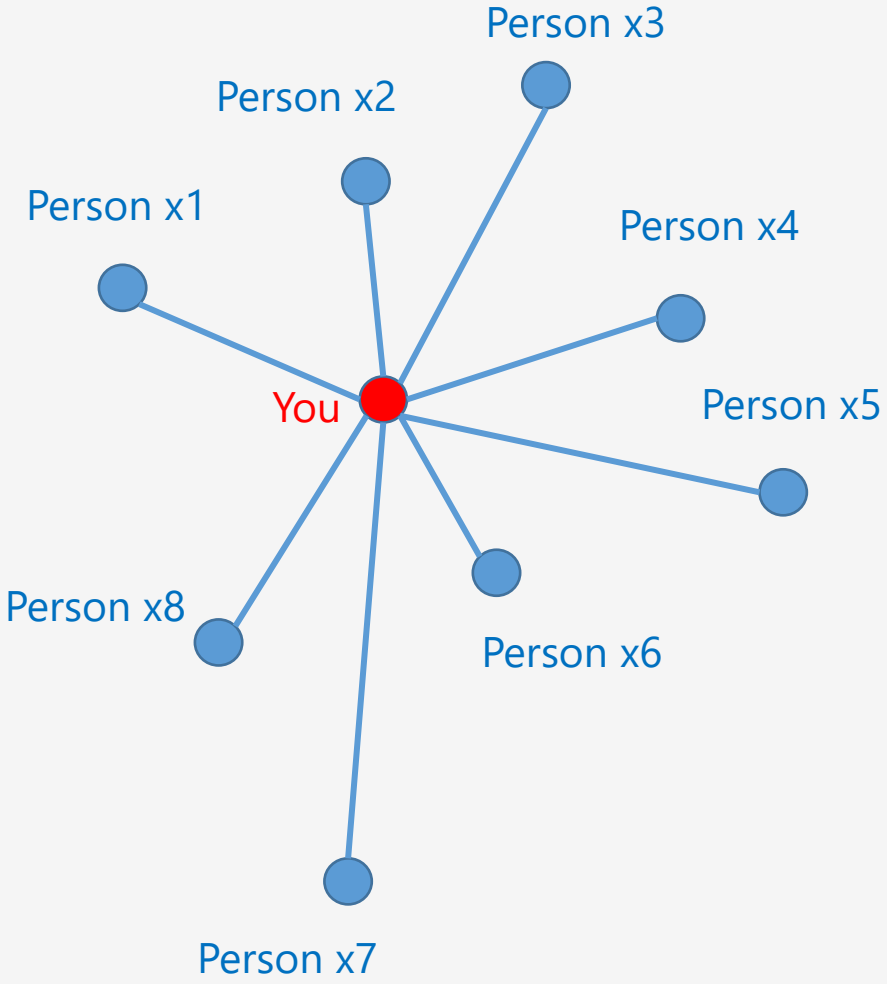
$$\text{s.t.} \qquad x \in \mathrm{R}^{\mathrm{n}}$$

$$\min_{x \in R^n} \; f(x)$$

- We will examine two applications:
  - **Application 1:** Fermat-Weber (Facility Location) Problem
  - **Application 2:** Least Squares

# Application #1: Fermat-Weber problem (aka, facility location problem)

# Application 1: Fermat-Weber problem

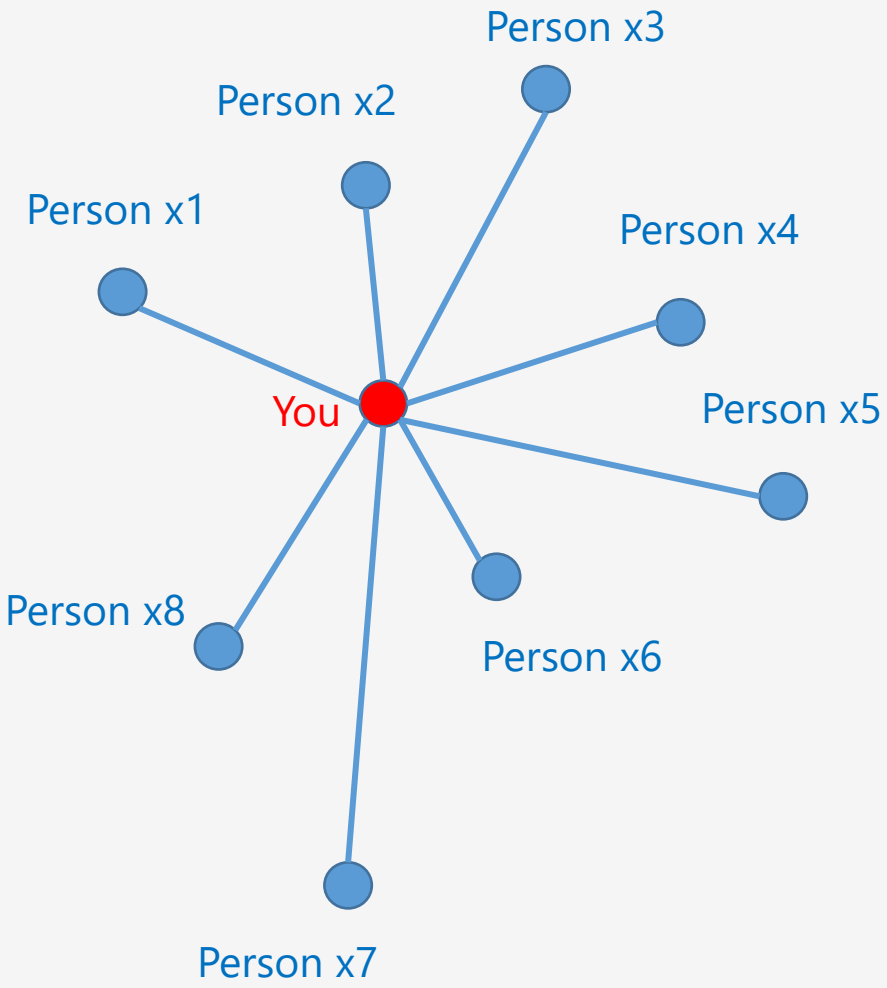



**Objective**: You want to find the location (of your new home) that is as close to all your loved ones as possible.

You know the coordinates of each $x_i \in R^2$ <u>(fixed points!)</u>
You are looking for your new home's coordinates, $y \in R^2$

The optimization you need to solve is**???**

# Application 1: Fermat-Weber problem




**Objective**: You want to find the location (of your new home) that is as close to all your loved ones as possible.

You know the coordinates of each $x_i \in R^2$
You are looking for your new home's coordinates, $y \in R^2$

The optimization you need to solve is:

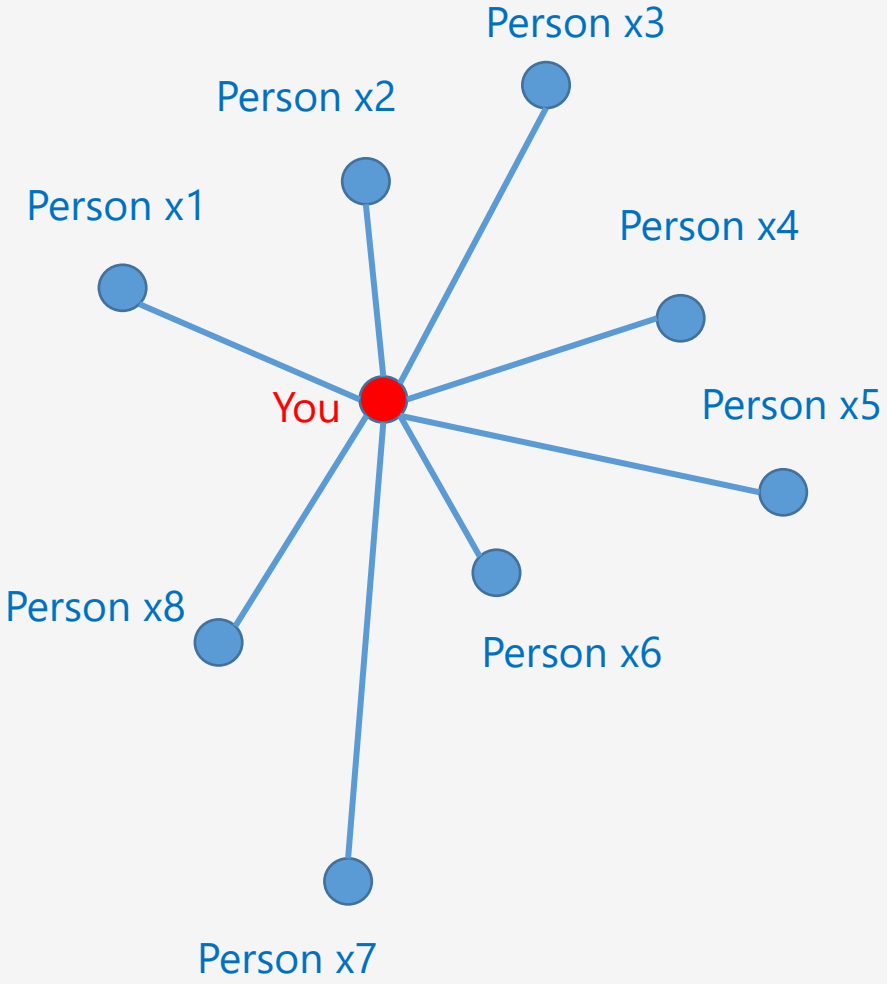
$$\min_{y \in R^n} \sum_{i=1}^{m} ||y - x_i||_2$$

where in this example we have: $n = 2$ and $m = 8$

If you have some preferred persons, you can include weights $w_i \in R, i = 1, \dots m$

$$\min_{y \in R^n} \sum_{i=1}^{m} w_i ||y - x_i||_2$$

# Application 1: Fermat-Weber problem




**Objective**: You want to find the location (of your new home) that is as close to all your loved ones as possible.

You know the coordinates of each $x_i \in R^2$
You are looking for your new home's coordinates, $y \in R^2$

The optimization you need to solve is:

$$\min_{y \in R^n} \sum_{i=1}^{m} ||y - x_i||_2$$

where in this example we have: $n = 2$ and $m = 8$

If you have some preferred persons, you can include weights $w_i \in R, i = 1, \dots m$

$$\min_{y \in R^n} \sum_{i=1}^{m} w_i ||y - x_i||_2$$

Fermat was a lawyer 😲

# Application #2: Least squares

# Application 2: Least Squares (LS)

- You are given a matrix $A \in R^{m \times n}$ and a vector $b \in R^m$ and asked to solve the unconstrained optimization problem

Sum of squares

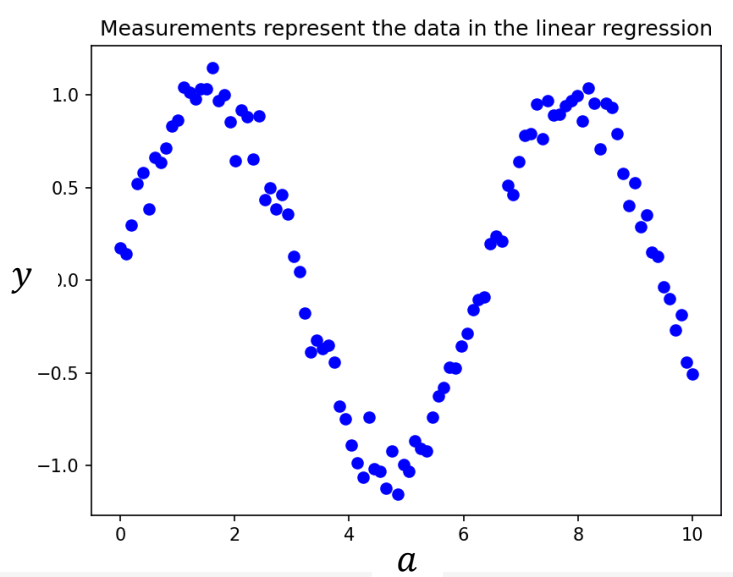
$$\min_{x \in R^n} \left|\left|Ax - b\right|\right|^2 = \min_x \sum_{i=1}^{m} \left(a_i^T x - b_i\right)^2$$

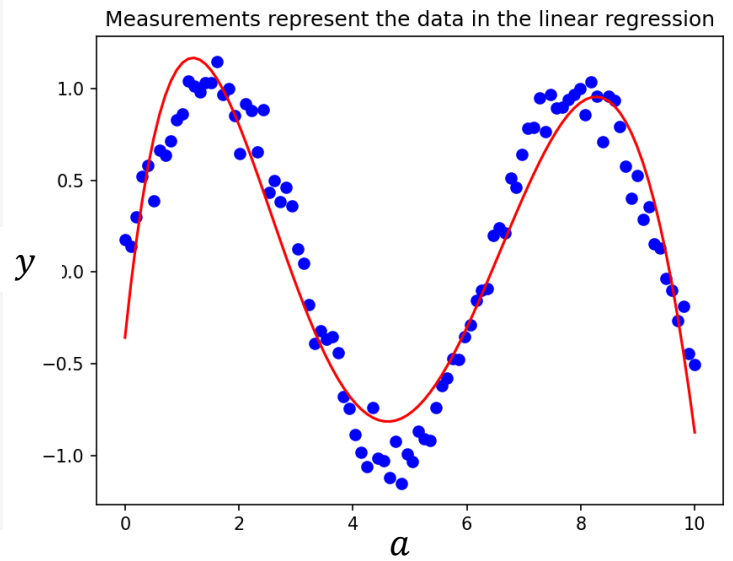
$$\mathrm{A} = \begin{pmatrix} \leftarrow & a_1^T & \rightarrow \\ & \dots & \\ \leftarrow & a_m^T & \rightarrow \end{pmatrix}$$, where $a_i^T$ is the $i$-th row of $A$, with $a_i \in R^n$

and $b = \begin{pmatrix} b_1 \\ \dots \\ b_m \end{pmatrix}$

Numerous applications in ML, Finance, Engineering, etc.

# Linear Regression (1st application of LS)




You have a set of **known** measurements $D = \{(a_i, y_i): i = 1, \dots, m\}$

You want to **fit** a **5th degree polynomial** to the data

$$\boxed{p_c(a) = c_5 a^5 + c_4 a^4 + c_3 a^3 + c_2 a^2 + c_1 a + c_0}$$

You are looking for the **optimal values** for the $c_5, c_4, c_3, c_2, c_1, c_0$ such that the polynomial $p(a)$ **fits all the data** as best as possible.
Hence your **decision variables** are $c_5, c_4, c_3, c_2, c_1, c_0$

The optimization problem we need to solve is:

Deviations

$$\min_{c_5, c_4, \dots, c_0} \sum_{i=1}^{m} (\overbrace{p_c(a_i) - y_i})^2$$

Or in expanded form

Linear in the $c$'s

$$\min_{c_5, c_4, \dots, c_0} \sum_{i=1}^{m} \left(c_5 a_i^5 + c_4 a_i^4 + c_3 a_i^3 + c_2 a_i^2 + c_1 a_i + c_0 - y_i\right)^2$$

# Linear Regression (1st application of LS)

- Write $\min_{c_5, c_4, \ldots, c_0} \sum_{i=1}^{m} \left(c_5 a_i^5 + c_4 a_i^4 + c_3 a_i^3 + c_2 a_i^2 + c_1 a_i + c_0 - y_i\right)^2$ as $\min_{x \in R^n} \left|\left|Ax - b\right|\right|^2$
  - You have to find $A, x$ and $b$
    - Tell me and I will write…
    - $A = ??$ $x = ??$ $b = ??$
- **Note**: the higher the degree ($n$) of the polynomial features ($a_i^n$) the larger the number of parameters ($c_i$) you have to compute. The resulting model fits the data better as $n$ increases. But you do not want to overdo it, because your model will overfit your data.

# Overdetermined systems of linear equations (2nd application of LS)

- You have some timeseries data $D = \{s_t : t = 1,2,\dots,n\}$.
  - For example, $s_t$ may be the closing price of a stock (AAPL, IBM. MSFT, etc.) at time (or day) $t$.
- You have observed that the price at time $t$ is exhibiting some relation with the prices of the **last 4 time points** (for example, the last 4 days)
  - That is, $s_t$ seems to have some (**unknown**) linear relation with the prices:
    - $s_{t-1}, s_{t-2}, s_{t-3}, s_{t-4},$ for all $t = 5,6,7,\dots,n$

# Overdetermined systems of linear equations (2nd application of LS)

- We are interested in **finding** (or ***learning***) that mysterious linear relationship
  - $\boldsymbol{s_t = c_1 s_{t-1} + c_2 s_{t-2} + c_3 s_{t-3} + c_4 s_{t-4}}$ for all $t = 5,6,7,\dots,n$
    - The unknown **decision variables** we are looking for are $c_1, c_2, c_3, c_4$
    - The stock prices, $s_{t-1}$, $s_{t-2}$, $s_{t-3}$, $s_{t-4}$, for all $t$, are known (**data**)
  - How can I express this **learning problem** as an **optimization problem**??

# Overdetermined systems of linear equations (2nd application of LS)

- We must find the **best** $c_1, c_2, c_3, c_4$ such that the following equations are satisfied as **accurately** as possible
  - $s_t = c_1 s_{t-1} + c_2 s_{t-2} + c_3 s_{t-3} + c_4 s_{t-4}$, **for** $\boldsymbol{t = 5, 6, 7, \ldots}$,
- We will model this learning problem as a least squares problem

$$\min_{x \in R^n} \left|\left|Ax - b\right|\right|^2 = \min_x \sum_{i=1}^{m} \left(a_i^T x - b_i\right)^2$$

  - **What are** $\boldsymbol{A} = ??, \boldsymbol{x} = ??, \boldsymbol{b} = ??$

# Overdetermined systems of linear equations (2nd application of LS)

$$A = \begin{bmatrix} s_4 & s_3 & s_2 & s_1 \\ s_5 & s_4 & s_3 & s_2 \\ s_6 & s_5 & s_4 & s_3 \\ \vdots & \vdots & \vdots & \vdots \\ s_{n-1} & s_{n-2} & s_{n-3} & s_{n-4} \end{bmatrix} \quad x = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} \quad b = \begin{pmatrix} s_5 \\ s_6 \\ s_7 \\ \vdots \\ s_n \end{pmatrix}$$

- **<u>Note</u>**: we have many more equations than unknowns. So, it is not possible to find $c = (c_1, c_2, c_3, c_4)$ that satisfies all equations.

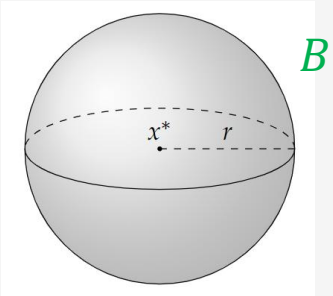
- The **least squares** optimization problem will give us the best $c^*$ that minimizes the norm, and hence approximately satisfies all equations

  - If for some reason, the objective function of the least squares problem is 0, then $c^*$ will satisfy all the equations exactly

# Optimality conditions for unconstrained optimization

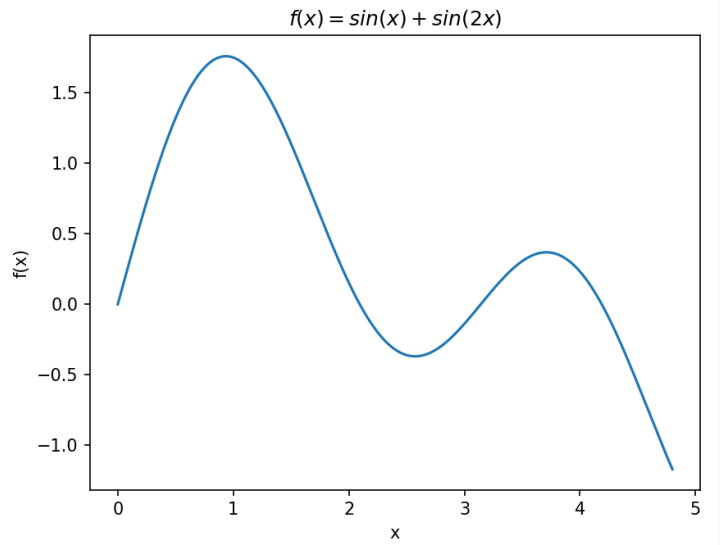
# Optimality Conditions for Unconstrained Optimization

Ball of radius $\delta > 0$ centered at $x^*$,




$B(x^*, \delta)$

- $f: R^n \to R$, and $x^* \in R^n$
- $x^*$ is a **local minimum** iff $\exists \delta > 0 \; st \; f(x^*) \leq f(x), \boxed{\forall x \; with \; ||x - x^*|| \leq \delta}$
- $x^*$ is a **strict local minimum** iff $\exists \delta > 0 \; st \; f(x^*) < f(x), \forall x \neq x^* \; with \; ||x - x^*|| \leq \delta$
- $x^*$ is a **global minimum** iff $f(x^*) \leq f(x), \forall x \in R^n$
- $x^*$ is a **strict global minimum** iff $f(x^*) < f(x), \forall x \neq x^* \in R^n$
- Analogous definitions exist of local/global maxima
- Question: is a global minimum a local minimum??
  - Which one is better?




**Exercise: Find all local/global min/max (which ones are strict?)**

# Optimality Conditions for Unconstrained Optimization

- **First Order <u>Necessary</u> Condition (FONC)**: if $x^*$ is an unconstrained local minimizer of $f: R^n \to R$ then $\nabla f(x^*) = 0$ *(Fermat 1600s)*

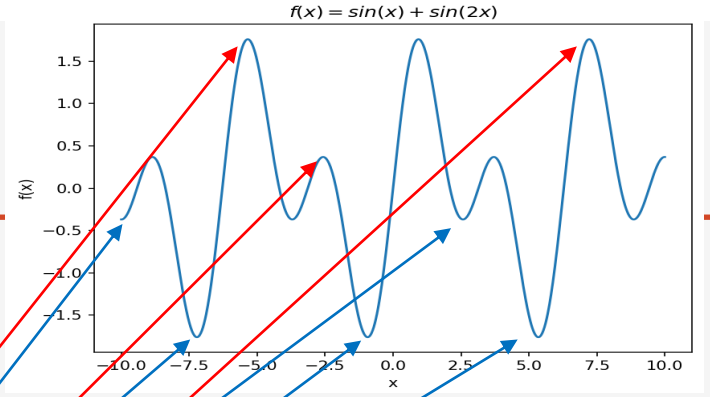
- **Second Order <u>Necessary</u> Condition (SONC):** if $x^*$ is an unconstrained local minimizer of $f: R^n \to R$ **then** $\nabla^2 f(x^*) \succcurlyeq 0$ (note that we also have $\nabla f(x^*) = 0$)

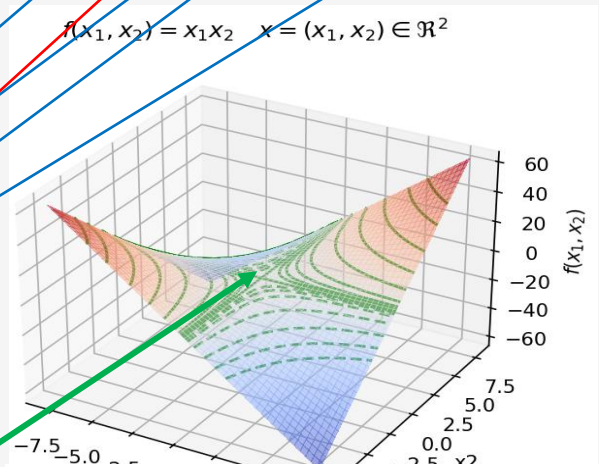
- **Second Order <u>Sufficient</u> Conditions (SOSC):** if $f$ is twice continuously differentiable and $\nabla f(x^*) = 0$ & $\nabla^2 f(x^*) \succ 0$ **then** $x^*$ is a *strict* local minimum of $f$.

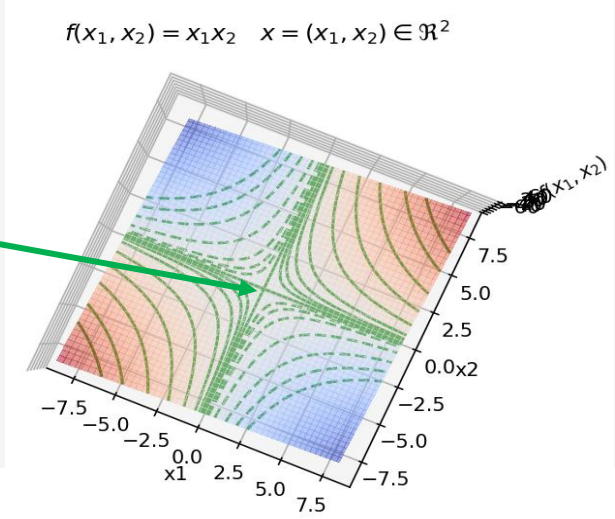
-

# Remarks for FONC

- The power of FONC is that it **reduces the search for local minima**. We can only consider those points $x \in R^n$ that make the gradient zero, i.e., $\nabla f(x) = 0$ (**<u>stationary</u>**)
- $\nabla f(x) = 0$ is a system of $n$ **nonlinear equations** because $x \in R^n$)
- All points $x$ that satisfy $\nabla f(x) = 0$ are **candidates** (may or may not be) for local minima (may also be maxima).
- A point $x \in R^n$ such that $\nabla f(x) = 0$ is called **critical point** if it is neither minimum or maximum



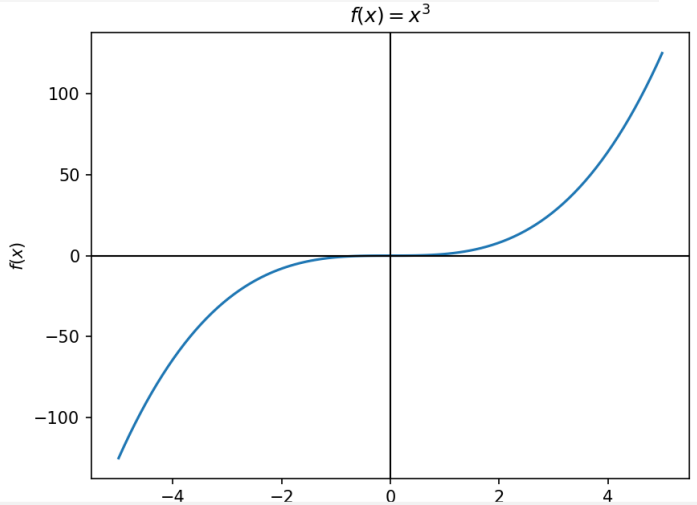



# Remarks on SOSC

- The conditions $\nabla f(x^*) = 0$ & $\nabla^2 f(x^*) \succcurlyeq 0$ are **NOT sufficient** for local optimality
  - **Example**:
- The condition $\nabla^2 f(x) \succ 0$ is **not necessary** for optimality
  - **Example**:

# Remarks on SOSC

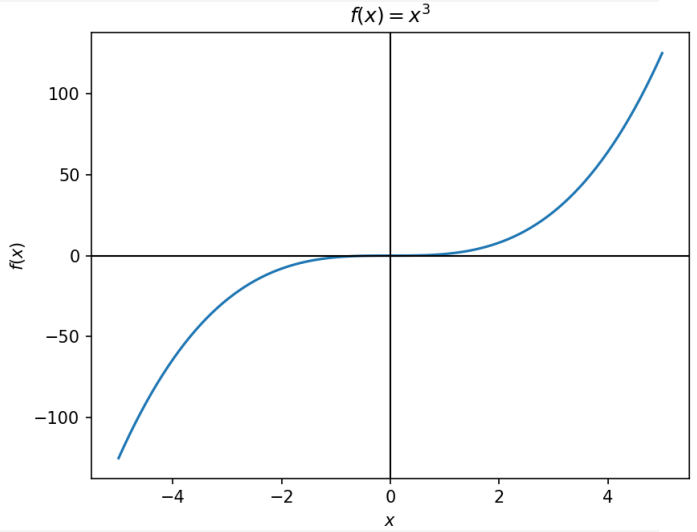
- The conditions $\nabla f(x^*) = 0$ & $\nabla^2 f(x^*) \succcurlyeq 0$ are **NOT sufficient** for local optimality
  - **Example**: $f(x) = x^3$,
    - $f'(0) = 0, \& \ f''(0) = 0$, but 0 is not a local minimum



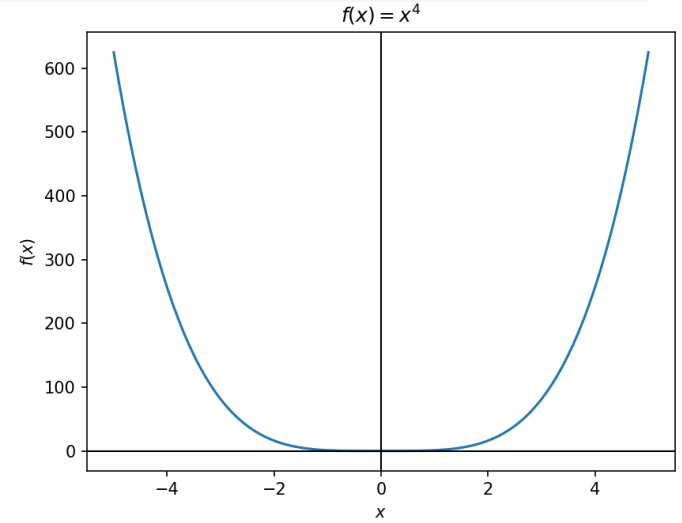

- The condition $\nabla^2 f(x) \succ 0$ is **not necessary** for optimality
  - **Example**:

# Remarks on SOSC

- The conditions $\nabla f(x^*) = 0$ & $\nabla^2 f(x^*) \succcurlyeq 0$ are **NOT sufficient** for local optimality
  - **Example**: $f(x) = x^3$,
    - $f'(0) = 0, \& \ f''(0) = 0$, but 0 is not a local minimum



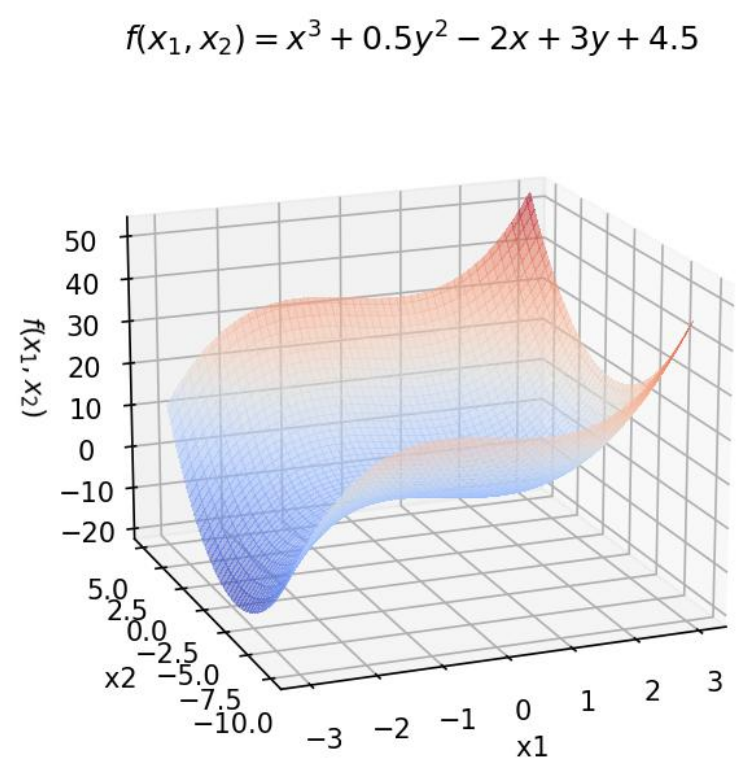

- The condition $\nabla^2 f(x) \succ 0$ is **not necessary** for optimality
  - **Example**: $f(x) = x^4$,
    - $f''(0) = 0$ and 0 is a strict global minimum

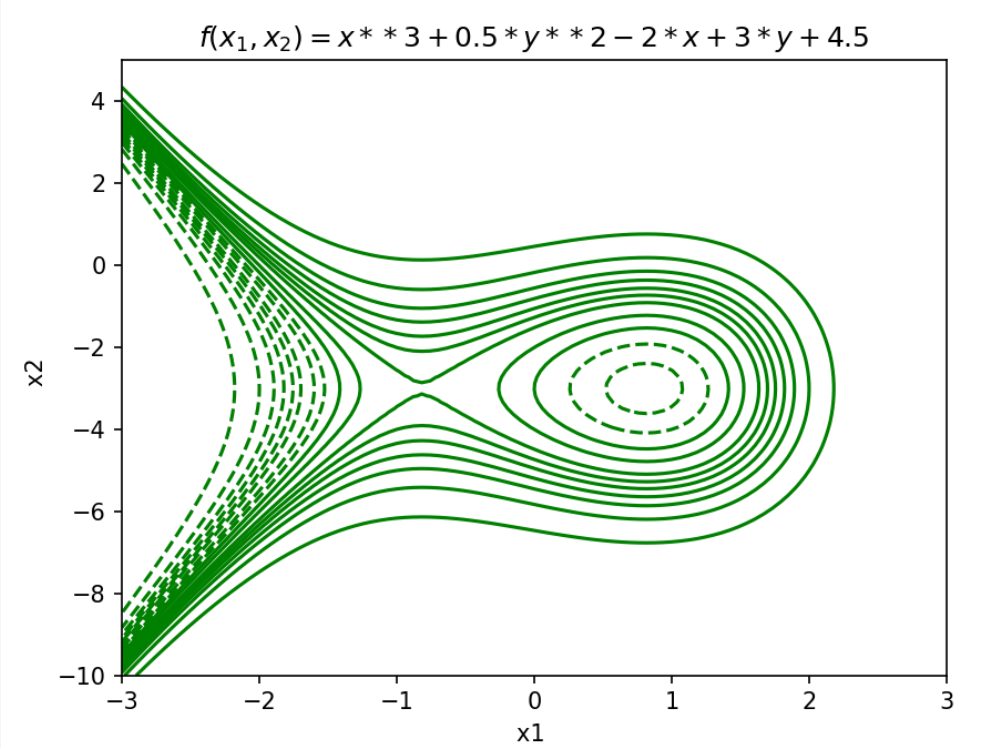


# Example/Exercise (you have 3 minutes)

- Find all minima and maxima of $f(x) = x_1^3 + 0.5x_2^2 - 2x_1 + 3x_2 + 4.5$

- Use the optimality criteria (FONC, SONC, SOSC)



# Optimality Conditions applied to Least Squares problem

- We will use the **Optimality Conditions** to find the optimal solution of **LS problem**
  - Recall: $\min_{x \in R^n} f(x) = \left|\left|Ax - b\right|\right|^2$
  - Also recall that the columns of $A \in R^{m \times n}$ are **linear independent**, and $b \in R^m$.
    - When is a group of $n$ vectors linear independent ??

# Optimality Conditions applied to Least Squares problem

- We will use the **Optimality Conditions** to find the optimal solution of **LS problem**
  - Recall: $\min_{x \in R^n} f(x) = \left|\left|Ax - b\right|\right|^2$
  - Also recall that the columns of $A \in R^{m \times n}$ are linear independent, and $b \in R^m$.
    - $A = [a_1\ a_2\ \ldots a_n]$, where $a_i \in R^m$ is the ***i*-th column** of $A$.
    - Linear independence of columns of $A$:
      - $\boxed{w_1 a_1 + w_2 a_2 + \cdots + w_n a_n = 0 \quad \Leftrightarrow \quad w_1 = w_2 = \cdots = w_n = 0}$

# Optimality Conditions applied to Least Squares

- We can write the **objective function** as follows (quadratic function):
  - $f(x) = (Ax - b)^T(Ax - b) = x^T A^T Ax - 2x^T A^T b + b^T b$
- Use **FONC**, $\nabla f(x) = 0 \Rightarrow 2A^T Ax - 2A^T b = 0 \Rightarrow A^T Ax = A^T b \Rightarrow \boxed{x = (A^T A)^{-1} A^T b}$
- Is $A^T A$ invertible matrix??

# Optimality Conditions applied to Least Squares

- We can write the **objective function** as follows (quadratic function):
  - $f(x) = (Ax - b)^T(Ax - b) = x^T A^T Ax - 2x^T A^T b + b^T b$
- Use **FONC**, $\nabla f(x) = 0 \Rightarrow 2A^T Ax - 2A^T b = 0 \Rightarrow A^T Ax = A^T b \Rightarrow \boxed{x = (A^T A)^{-1} A^T b}$
- Is $A^T A$ invertible matrix??
- **Recall: any square matrix $\boldsymbol{B} \in \boldsymbol{R}^{n \times n}$ is invertible $\Leftrightarrow (\boldsymbol{Bx} = \boldsymbol{0} \Rightarrow \boldsymbol{x} = \boldsymbol{0})$**
- Applying this to $A^T A \in R^{n \times n}$ and we have:

*Columns of A are linear independent*

$$A^T Ax = 0 \Rightarrow x^T A^T A\, x = x^T 0 \Rightarrow (Ax)^T (Ax) = 0 \Rightarrow \left|\left|Ax\right|\right|^2 = 0 \Rightarrow Ax = 0 \Rightarrow \sum_{i=1}^{n} a_i x_i = 0 \Rightarrow x = 0 \Rightarrow (A^T A)^{-1} \text{exists}$$

# Optimality Conditions applied to Least Squares

- Hence $x = (A^T A)^{-1} A^T b$ is a **candidate solution** of the least squares problem.
- What do I need to do to show it is a minimum?? Check the Hessian
  - $\nabla^2 f(x) = \nabla(A^T A x) = A^T A$
- How can we show that $\nabla^2 f(x)$ is PSD??
- Yes, by using the definition of PSD
  - Choose any $x \in R^n$ and then: $x^T (A^T A) x = (Ax)^T Ax = ||Ax||^2 \geq 0 \Rightarrow A^T A \succcurlyeq 0$
  - But the columns of $A$ are linear independent $\Rightarrow$ $Ax = 0 \Rightarrow x = 0$.
  - Hence since $x = (A^T A)^{-1} A^T b \neq 0$ we have $||Ax||^2 > 0 \Rightarrow A^T A \succ 0$, and $\boldsymbol{x = (A^T A)^{-1} A^T b}$ **is a strict local minimum.**

# Optimality Conditions applied to Least Squares

- Hence $x = (A^T A)^{-1} A^T b$ is a **candidate solution** of the least squares problem.
- What do I need to do to show it is a minimum?? Check if the Hessian is PD
  - $\nabla^2 f(x) = \nabla(A^T A x) = A^T A$
- How can we show that $\nabla^2 f(x)$ is PD??

# Optimality Conditions applied to Least Squares

- Hence $x = (A^T A)^{-1} A^T b$ is a **candidate solution** of the least squares problem.
- What do I need to do to show it is a minimum?? Check if the Hessian is PD
  - $\nabla^2 f(x) = \nabla(A^T A x) = A^T A$
- How can we show that $\nabla^2 f(x)$ is PD??
- **Yes, by using the definition of PD**
  - Choose any $x \in R^n$ and then: $x^T (A^T A) x = (Ax)^T Ax = \left|\left|Ax\right|\right|^2 \geq 0 \Rightarrow A^T A \succcurlyeq 0$
  - But the columns of $A$ are linearly independent $\Rightarrow$ $Ax = 0 \Rightarrow x = 0$.
  - Hence since $x = (A^T A)^{-1} A^T b \neq 0$ we have $||Ax||^2 > 0 \Rightarrow A^T A \succ 0$, and $\boldsymbol{x = \left(A^T A\right)^{-1} A^T b}$ **is a strict local minimum.**